TT-R-771B
November 12, 1968
SUPERSEDING
Int. Fed. Spec. TT-R-00771A(GSA-FSS)
Jan. 15, 1962 and
Fed. Spec. TT-R-771
Oct. 2, 1953

FEDERAL SPECIFICATION

RUBBING COMPOUND, LACQUER

This specification was approved by the Commissioner Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers rubbing compound for use on lacquered surfaces to produce a smooth and high gloss finish.

1.2 Classification. Rubbing compounds shall be of the following types, as specified (see 6.2):

Type I - For hand rubbing
Type II - For machine use

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

Federal Specification:

QQ-A-250/5 - Aluminum Alloy Alclad 2024, Plate and Sheet.
QQ-S-698 - Steel, Sheet and Strip, Low-Carbon.
TT-P-664 - Primer coating Synthetic Rust Inhibiting Lacquer Resisting.
CCC-C-458 - Cloth, Flannel, Cotton.
PPP-C-96 - Cans, Metal 28 Gage and Lighter.
PPP-P-704 - Pails, Metal: (Shipping Steel, 1 through 12 gallon).

Federal Specification:

Fed. Std. No. 123 - Marking for Domestic Shipment.
Fed. Test Method Std. No. 141 - Paint, Varnish, Lacquer and Related Materials; Methods of Inspection, Sampling and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Wash.)

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

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Military Standard:

MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurment functions should be obtained from the procuring activity or as directed by the contracting officer)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Uniform Classification Committee:

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

National Classification Board, Inc.

National Motor Freight Classification Rules.

(Application for copies should be addressed to the National Classification Board, Inc., 1616 P St., N.W., Washington, D.C. 20036.)

3. REQUIREMENTS.

3.1 Composition.

3.1.1 Material. The compound covered by this specification shall be a paste mixture containing a finely divided abrasive.

3.1.2 Toxicity. The compound shall contain no material which would constitute medical hazard to personnel when used as specified herein, and must not produce obnoxious vapors in such concentrations as to become an annoyance or a toxic hazard to personnel.

3.1.3 Chlorinated hydrocarbons. The compound shall contain no chlorinated hydrocarbons with the following exceptions:

(1) Dichloromethane may be used.
(2) Trichlorethylene and perchlorethylene may be used in low concentrations.

Compounds containing these solvents will be evaluated for toxicity prior to recommendation for acceptance.

3.2 Form. The compound shall be furnished as a paste.

3.3 Chemical attack. The compound shall not chemically attack a lacquered surface when tested as specified in 4.3.

3.4 Corrosiveness. The compound shall not discolor or attack aluminum-clad aluminum-alloy surface when tested as specified in 4.3.

3.5 pH value. The pH value of the compound, when determined as specified in 4.3 shall not exceed 10.7.

3.6 Flash point. The flash point of the compound shall be not less than 60° C. (140°F.) when determined as specified in 4.3.

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3.7 Rubbing efficiency. The sum of the average loss in weight of each of two duplicate panels, after the rubbing efficiency test described in 4.3.5, shall be as follows:

Type I - between 44 and 75 milligrams.
Type II - between 10 and 23 milligrams.

3.8 Appearance of panel. The lacquer surface after the rubbing efficiency test specified in 4.3.5 shall be smooth, free from orange peel and scratches, and shall be sufficiently thick to completely hide the prime undercoat.

3.9 Workmanship. The ingredients shall be homogeneously blended and processed as required to produce a rubbing compound which is stable and not subject to abnormal change with age in a sealed container.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling and inspection. Unless otherwise specified, sampling and inspection shall be in accordance with the applicable Methods 1011, 1021 and 1031 of Fed. Test Method Std. No. 141.

4.3 Test Procedures.

4.3.1 The following tests shall be conducted in accordance with Federal Test Method Std. No. 141.

Test	Method No.
Flash Point	4293

4.3.2 Chemical attack. Vapor degrease two 3 inch x 6 inch panels of 20-gage cold rolled 1020 steel strip conforming to QQ-S-698 and spray with a coat of primer conforming to TT-P-664. Allow to air-dry for 4 hours. Spray 2 coats of olive drab automotive lacquer conforming to test lacquer in table I of 4.3.5.2 and allow to air-dry for 24 hours. Coat both panels with the rubbing compound in accordance with the manufactuer's instructions and allow to stand in room temperature 25° C. (77°F.) for 24 hours. Wipe one-half of each panel clean and examine the lacquer surface for any sign of chemical attack. Allow the panels to stand an additional 24 hours, and wipe the remaining portions clean. Examinie the lacquer surface for any sign of chemical attack.

4.3.3 Corrosiveness. Coat two 1 inch x 6 inch panels of aluminum alloy alclad conforming to QQ-A-250/5 with the rubbing compound in accordance with the manufacturer's instructions and allow to stand at room temperature 25° C. (77°F.) for 24 hours. Wipe one-half of each panel clean and examine the cleaned surfaces for corrosive attack, staining, or discoloration. Allow the panels to stand an additional 24 hours and wipe the remaining portions clean. Examine the cleaned surfaces for corrosive attack, staining, or discoloration.

4.3.4 pH value. Place a 10 plus or minus 0.5 gram sample of the rubbing compound in a 100-ml. beaker and add 75 ml. of distilled water. Stir the mixture vigorously for one minute and determine the pH value electrometrically to the nearest o.1 pH unit.

4.3.5 Rubbing efficiency.

4.3.5.1 Rubbing block. The rubbing surface of the block shall be 4 inches long by 3 inches wide and designed so that a strip of flannel may be securely clamped to and stretched taut over the block in order to completely cover the bottom rubbing area. Provision shall be made for the attachment of lead

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weights to the top of the block in an evenly apportioned manner to give total weights of 8 and 15 pounds. The block shall be operated in a level, smooth, reciprocating motion through a length of 2 inches in a direction parallel to the length of the block. The block shall operate at a speed of 40 strokes per minute, a stroke being considered as one complete cycle from starting position of block and return to starting position.

4.3.5.2 Panels. Four steel panels 3 by 5 by 0.125 inch of 1020 steel, one face of which shall be machined level, shall have the level face coated with one coat of primer, TT-P-664, and two coats of olive drab automotive lacquer conforming to test lacquer in table I, so that the total dry film thickness is 1.5 to 1.7 mil. After air drying for 10 days the panels shall be washed vigorously with a mild soap solution, rinsed, and dried thoroughly and weighed to the nearest milligram.

TABLE I. Test lacquer

Ingredient	Percent by weight
Yellow dispersion (1)	8.0
Orange dispersion (2)	8.0
White dispersion (3)	2.0
Black dispersion (4)	6.0
Nitrocellulose RS 1/2 second (65 percent in ethanol)	7.0
Non-drying phthalic alkyd resin (65 percent in xylol) (5)	12.5
Blown castor oil (6)	1.6
Dioctyl phthalate	1.6
Ethyl acetate	6.6
Butyl acetate	13.3
Ethyl alcohol	4.3
Butyl alcohol	7.7
Toluol	21.4

	Percent by weight
(1) Shall consist of the following:	
Chrome yellow lemon	50.0
RS 1/2 second nitrocellulose	11.0
Ethyl alcohol	4.5
Toluol	20.5
Ethyl acetate	14.0
(2) Shall consist of the following:	
Chrome orange medium	60.0
RS 1/2 second nitrocellulose	17.0
Ethyl alcohol	7.0
Ethyl acetate	6.5
Toluol	9.5
(3) Shall consist of the following:	
Titanium dioxide	60.0
RS 1/2 second nitrocellulose	8.0
Ethyl alcohol	3.5
Ethyl acetate	16.0
Toluol	12.5
(4) Shall consist of the following:	
Carbon black	8.5
RS 1/2 second nitrocellulose	20.0
Ethyl alcohol	8.0
Ethyl acetate	25.0
Toluol	38.5
(5) Shall contain the following:	
Non-drying phthalic alkyd resin;	
Phthalic anhydride	percent 35
Castor oil	percent 45

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TABLE I. Test lacquer (con.)

Ingredient	Percent by weight
(6) Shall have the following characteristics:	
Viscosity (Gardner-Hold)	(1)
Acid number	10 - 15
Iodine number	50 - 60
Saponification number	201
Color (Gardner Color Standard)	11

(1) heavier than Z-6.

4.3.5.3 Procedure. A weighed panel, prepared as specified above, shall be held securely in position beneath the rubbing block with the 5 inch dimension parallel to the length of the block. A fresh strip of flannel, CCC-C-485, type III, class 1, shall be clamped on the rubbing block. Weights shall be added to a total rubbing block weight of eight pounds. Two ml. of the rubbing compound shall be measured by means of a calibrated plunger assembly, care being taken to eliminate air pockets. The rubbing compound shall be spread, using a spatula, as evenly as possible over the portion of the flannel which will contact the panel. The rubbing block shall be placed in position and moved through 100 strokes. Remove the rubbing block and discard the flannel. Wash the panel thoroughly using a mild soap solution and a soft bristle brush, dry thoroughly and weigh. Repeat the above procedure on a second panel, using fresh flannel and rubbing compound. Subject the third and fourth panels to the entire procedure outlined above except that total rubbing block weight of 15 pounds and only 40 strokes shall be used. Fresh flannel and rubbing compound shall be used for each panel. The first and second panels tested above shall be paired as one set of two panels, and the third and fourth panels shall be considered as the other set to two panels.

4.3.5.4 Results. The loss in weight of each panel shall be calculated in milligrams. The average loss in weight of each set of two panels shall be calculated. The sum of the two averages shall meet the requirements of 3.7.

4.3.6 Appearance of panel. Examine each panel after the completion of the rubbing efficiency test for compliance with the requirements of 3.8.

5. PREPARATION FOR DELIVERY.

5.1 Packaging. Packaging shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A. The rubbing compound shall be furnished in 1-quart, 5-pound, or 25-pound quantities, as specified, (see 6.2) and filled into 1-quart or 1-gallon cans conforming to PPP-C-96, or in 5-gallon pails conforming to PPP-P-704, as applicable.

5.1.2 Level B. The rubbing compound shall be furnished in 1-quart, 5-pound or 25-pound quantities as specified, (see 6.2) and filled into 1-quart, or 1-gallon cans conforming to PPP-C-96, type V, class 2 or in 5-gallon pails conforming to PPP-P-704, type I or II, class optional.

5.1.3 Level C. The rubbing compound, shall be packaged to assure adequate protection against damage during shipment from the supplier to the initial destination.

5.2 Packing. Packing shall be level A, B, or C as specified, (see 6.2).

5.2.1 Level A. The packaged rubbing compound shall be packed in accordance with the level A requirements of the appendix to PPP-C-96, except that 25 pound quantities in metal parts conforming to PPP-P-704 shall require no further packing for shipment.

5.2.2 Level B. Twelve 1-quart cans; or four 1-gallon cans of 5-pound quantities of rubbing compound shall be packed in accordance with the level B requirements of the appendix to PPP-C-96. Twenty-five pounds filled into a PPP-P-704 five gallon pail shall not require any further packing.

5.2.3 Level C. The rubbing compound, in quantities as specified (see 6.2), packaged as specified in 5.1, shall be packed in containers to assure carrier acceptance and safe arrival at destination in compliance with Uniform Freight Classification Rules or National Motor Freight Classification Rules.

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5.3 Marking. In addition to any special markings required by procurement document the interior packages and the shipping containers shall be marked in accordance with Fed. Std. No. 123 for civilian agencies and MIL-STD-129, for military agencies as applicable.

6. NOTES.

6.1 Intended use. The rubbing compound covered by this specification is intended for use as a hand or machine compound on lacquered surfaces to remove surface scratches, orange peel, spray pattern, and overspray as well as for cleaning severely weathered finishes before waxing.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement document:

(a) Title, number, and date of this specification.
(b) Type required (see 1.2).
(c) Selection of applicable levels of packaging and packing required (see 5.1 and 5.2).
(d) Size of container required (see 5.).

6.3 The material shall be purchased by weight. The weight of the rubbing compound is approximately ten pounds per gallon. The one quart container is preferable.

MILITARY CUSTODIANS:

Review activities:

Army - MR, WC
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User activities:

Army - MI, ME, AT
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Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain copies of documents referenced herein. Price ents each.

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